Huy Le

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EDUCATION

University of Pittsburgh — Pittsburgh, PA
Doctor of Philosophy in Statistics — Expected August 2026
Dissertation defended; Advisor: Chris McKennan, PhD

University of California, Berkeley — Berkeley, CA
Master of Arts in Statistics — 2019

Oregon State University — Corvallis, OR
Bachelor of Science in Mathematics, Minor in Actuarial Science — 2018
GPA: 4.0/4.0, Summa Cum Laude, Award for Outstanding Performance in Mathematics, Oregon State University

SKILLS

Programming: Python, R, SQL, MATLAB, Git, Bash/Shell
Methods: Probability, Stochastic Processes, Statistical Inference, Time Series, Optimization, Bayesian Modeling, Empirical Bayes, Gaussian Processes, Simulation Studies, Model Validation, Machine Learning

RESEARCH EXPERIENCE

Graduate Research Assistant — June 2020 – Present
Department of Statistics, University of Pittsburgh — Pittsburgh, PA

Bayesian Spatiotemporal Modeling for Longitudinal High-Dimensional Data

- Developed Bayesian spatiotemporal models in R and Python for high-dimensional longitudinal data, combining latent Gaussian processes, spike-and-slab priors, and covariance estimation to identify persistent and time-varying signals.
- Built an end-to-end inference pipeline using EM-style estimation, posterior simulation, and region-level error-control procedures for statistically calibrated signal detection.
- Designed simulation studies to evaluate calibration, power, false discovery control, and robustness under varying signal, noise, dependence, and heterogeneity regimes.
- Applied the framework to longitudinal cohort data with 750,000+ ordered features, translating feature-level summaries into interpretable region-level inference.

Spatial Empirical Bayes Modeling for Heterogeneous Correlated Data

- Developed empirical Bayes models for heterogeneous spatial dependence, allowing local inference to adapt when correlation structure varies across ordered features.
- Implemented scalable approximation and optimization methods, including expectation propagation, importance sampling, pseudo-likelihood estimation, and mixture optimization.
- Evaluated model behavior against standard region-detection baselines through simulation studies focused on calibration, localization, and posterior uncertainty quantification.
- Produced posterior summaries, diagnostic checks, and reproducible analyses to support statistically grounded interpretation of high-dimensional correlated data.

PROFESSIONAL EXPERIENCE

Quantitative Modeling Intern (PK/PD) — June 2023 – August 2023
AbbVie — South San Francisco, CA

- Developed and validated 6 ODE-based models in MATLAB to support drug feasibility and development-stage decision making.
- Built and deployed R Shiny simulation applications for real-time exploration of concentration–response relationships.
- Compared model predictions against experimental data to assess predictive performance and support candidate evaluation.
- Implemented Git-based version control to improve reproducibility and collaboration across modeling workflows.

Teaching Assistant — June 2020 – Present
Department of Statistics, University of Pittsburgh — Pittsburgh, PA

- Primary instructor or teaching assistant for probability, mathematical statistics, and introductory/intermediate statistics courses; communicated technical ideas to students with varied quantitative backgrounds.

PROJECTS

Spatial Risk Forecasting and Route Optimization — January 2019 – May 2019
University of California, Berkeley — Berkeley, CA

- Built a crime-risk forecasting model in R using zero-inflated negative binomial regression for overdispersed, zero-inflated count data.
- Designed a route-optimization algorithm that combined spatial risk forecasts with public safety data to identify statistically safer routes.